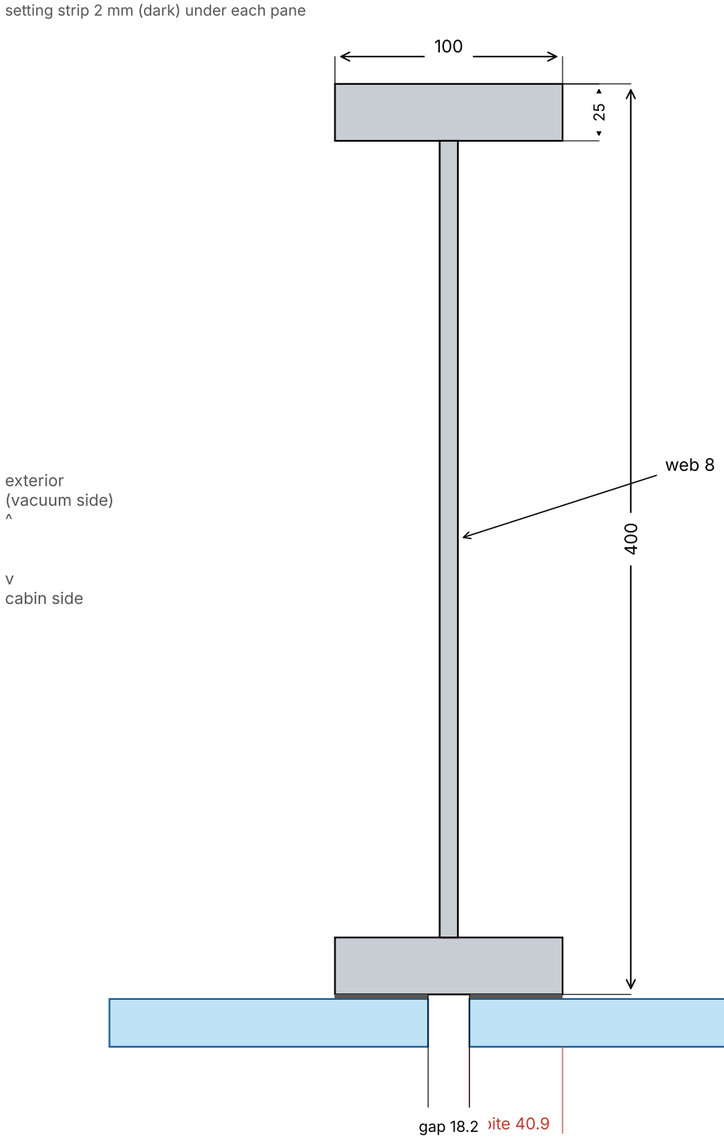
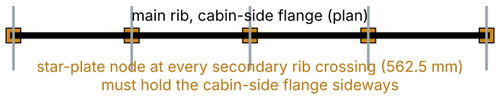
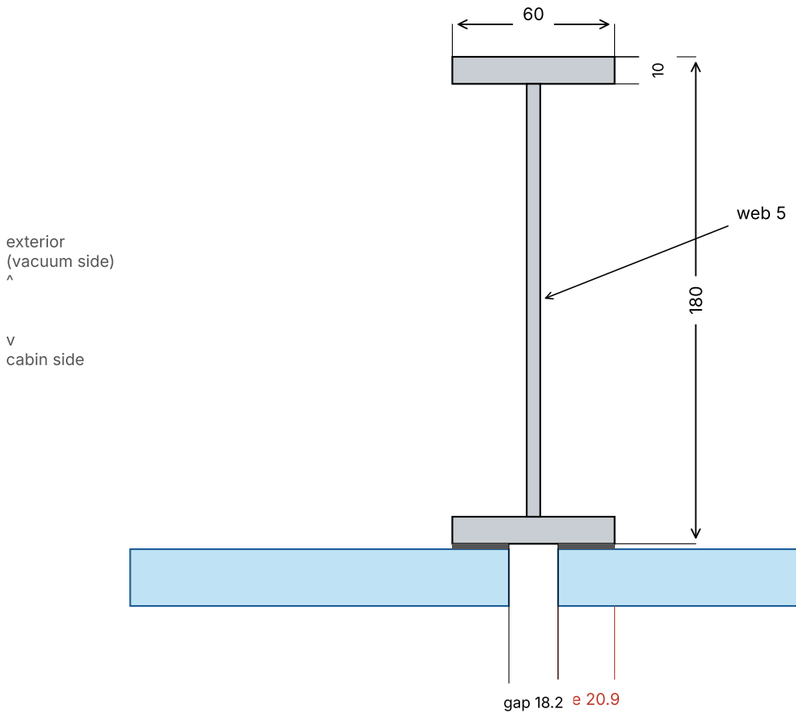
Main rib, option B: I 400 × 100 × 25, web 8
P-04 (hexagon), P-04A (pentagon), 6061-T6, 21.1 kg/m



21 mm BOROFLOAT 33 pane, 531 mm cut, cut 9.09 mm short of the rib centre line



Secondary rib: I 180 × 60 × 10, web 5
custom extrusion, 6061-T6, 5.4 kg/m

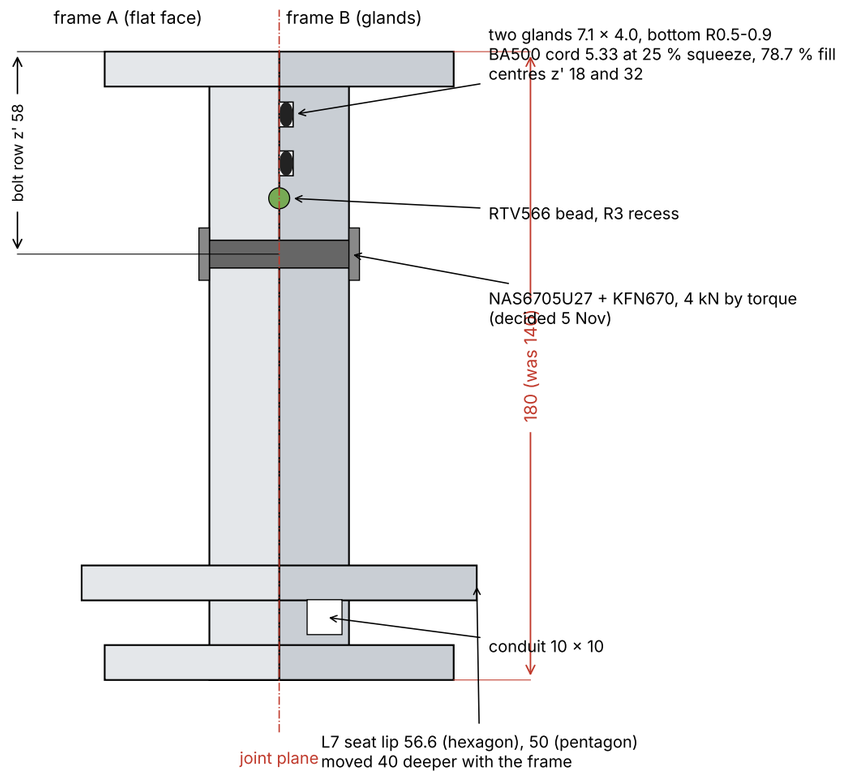


21 mm BOROFLOAT 33 pane, 531 mm cut, cut 9.09 mm short of the rib centre line

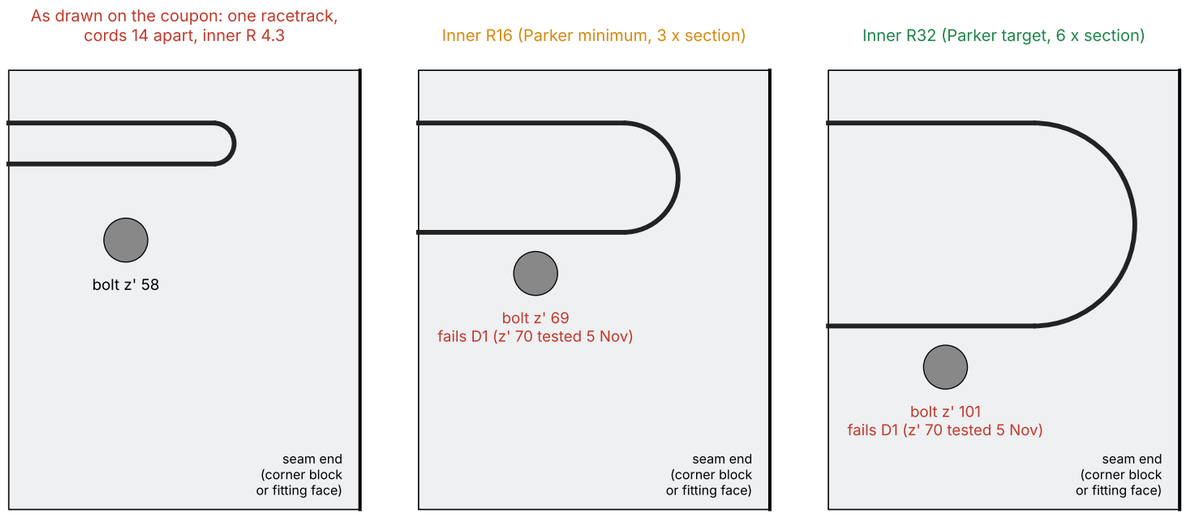
Rule checks: glass bite >= 20 mm on every seat (secondary 20.9, main 40.9); pane edge clears the web by 6.6 (secondary) and 5.1 mm (main).
Option A (I 450 × 150 × 25, web 10, parts P-10/P-10A) is superseded, kept for the record only.
Main rib B yields at 1.96 atm; at the 0.8 atm cabin the ultimate is 1.67 atm.
Open glass straight on with these ribs and a 60 mm frame band: 60.9 % of a hexagon (k 4 grid, exact drawing).
REQUIREMENT (survival check, 4 Nov): the main rib fails lateral-torsional buckling if unbraced over 2.25 m (0.79 of the load it must carry), and has 8.4 times margin if braced at 562 mm. The star-plate node at each secondary rib crossing must restrain the cabin-side flange sideways; carry this note onto RE-05 and RE-06.
LAUNCH NOTE: packed outer face down at 6 g, each pane pulls off its seat with about 0.35 kN. Pack the panels outer face up, or rate the pane clamps for 6 g plus random vibration (OPEN). Carry this onto the pane and seat sheet RE-04 and the packing plan (TRUTH 8.5.5).

HCSA RF-01 Main rib B and secondary rib with panes	
Rev F working baseline, 5 Nov 2026 (supersedes the Rev E draft sheet)	Dimensions in mm
Status: main rib B 4 Nov, secondary 2 Nov, pane 21 mm 5 Nov: decided	Redrawn from hcsa_params_revF.json
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining until checked

Section through a seam, normal to the joint plane (schematic, miter not shown)



Seal loop ends on the joint face (exterior at the top, z' down). Which layout we build waits on Parker's 'two loops' question.

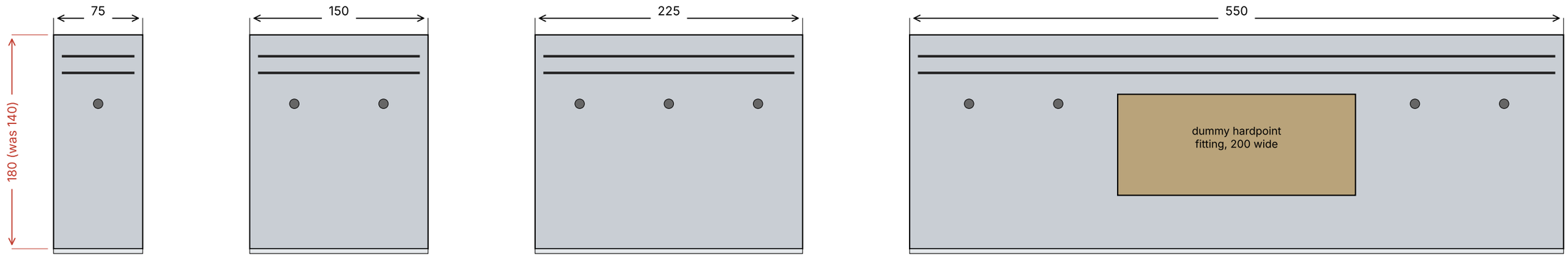


What the check says

- If each seam's two cords are one racetrack joined at the seam ends, the ends turn at inner R 4.3 mm. That breaks Parker's rule (R16 minimum).
- A racetrack with an inner R16 end needs its legs 37.3 mm apart, which pushes the bolt row from z' 58 to z' 70 or more (z' 102 for R32).
  The D1 run of 5 Nov says no: at z' 70 the seal gap is 0.111 mm at 2.48 kN per bolt, on the 0.11 limit (3.75 kN at z' 58). The bolt row stays at z' 58.
- If each cord is its own closed loop around the whole panel (Rev B intent), the turns are at the vertices, 60 deg on a hexagon, 72 deg on a pentagon,
  through the corner block and the O10 corner seal. That corner needs CAD of the block; it is not checked here.
- Next: draw the loop path through the corner block in Fusion
  (turn the loop round the end bolt, or widen the frame flange locally at the panel corners), after Parker answers 'two loops'.

HCSA RF-02 Seam frame 180, gland, seal loop corners	
Rev F working baseline, 5 Nov 2026 (supersedes the Rev E draft sheet)	Dimensions in mm
Status: frame 180, gland 7.1, bolt 4 kN: decided 5 Nov; loop corners open	Redrawn from hcsa_params_revF.json
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining until checked

Elevation on the joint face: frame depth 180, two cord lines at z' 18 and 32, bolt row at z' 58 (kept: z' 70 failed D1, 5 Nov)



RE-01-75
1 bolt at 75 pitch

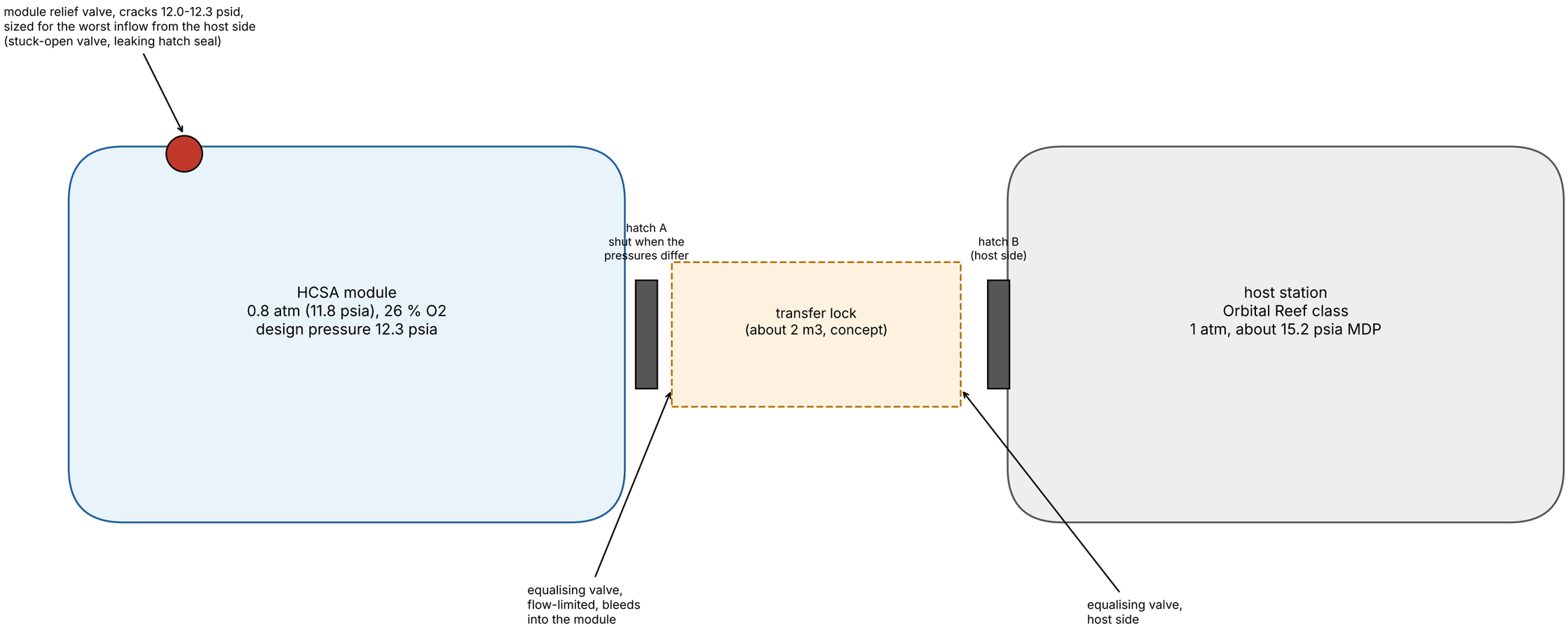
RE-01-150
2 bolts at 75 pitch

RE-01-225
3 bolts at 75 pitch

RE-02-550 with a dummy hardpoint
4 bolts at 75 pitch

Changes from the Rev E draft sheets RE-01 and RE-02:
- Frame depth 140 → 180 on every T1 frame, so the test matches the station frame (decided 5 Nov). Stock: 6061-T651 plate; check the blank with the shop.
- Glands 7.1 × 4.0 (decided 5 Nov). Cord ends at the slice ends follow RF-02: no tight racetrack end; use R16 minimum or dam the ends.
- Bolts NAS6705U27 + KFN670 at 4 kN by torque (decided 5 Nov), row at z' 58.
- The 550 article carries the dummy hardpoint of RE-03 rev B (no bolt inside the fitting: 2 + 2 bolts, 31.25 mm off its faces), pulled through pinned lugs.
- The pull blocks of RE-03 rev B keep their interface; their bolt holes move with the 40 mm deeper frame.
- The Instron adapter waits on Jerry's measurements.

HCSA RF-03 T1 test articles at 180 deep	
Rev F working baseline, 5 Nov 2026 (supersedes the Rev E draft sheet)	Dimensions in mm
Status: depth 180, gland 7.1, bolt 4 kN: decided 5 Nov	Redrawn from hcsa_params_revF.json
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining until checked



Rules this concept has to keep (from the 3 Nov lower-pressure note):
1. The module has its own relief valve, cracking at about 12.0-12.3 psid, big enough for the worst inflow from the 1 atm side.
2. Hatch A stays shut whenever the pressures differ, held by the pressure difference and by an interlock.
3. The lock does the stepping through flow-limited valves: about 6-10 minutes at 0.3-0.5 psi per minute, slowest on the way back up to 1 atm.
4. The module never takes its air from the host. Volumes, valve sizes and the hatch are not sized; the operator has to agree to a hatch-shut interface.

HCSA RF-04 Docking transfer lock, concept	
Rev F working baseline, 5 Nov 2026 (supersedes the Rev E draft sheet)	Dimensions in mm
Status: concept, Rick to agree	Redrawn from hcsa_params_revF.json
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining until checked